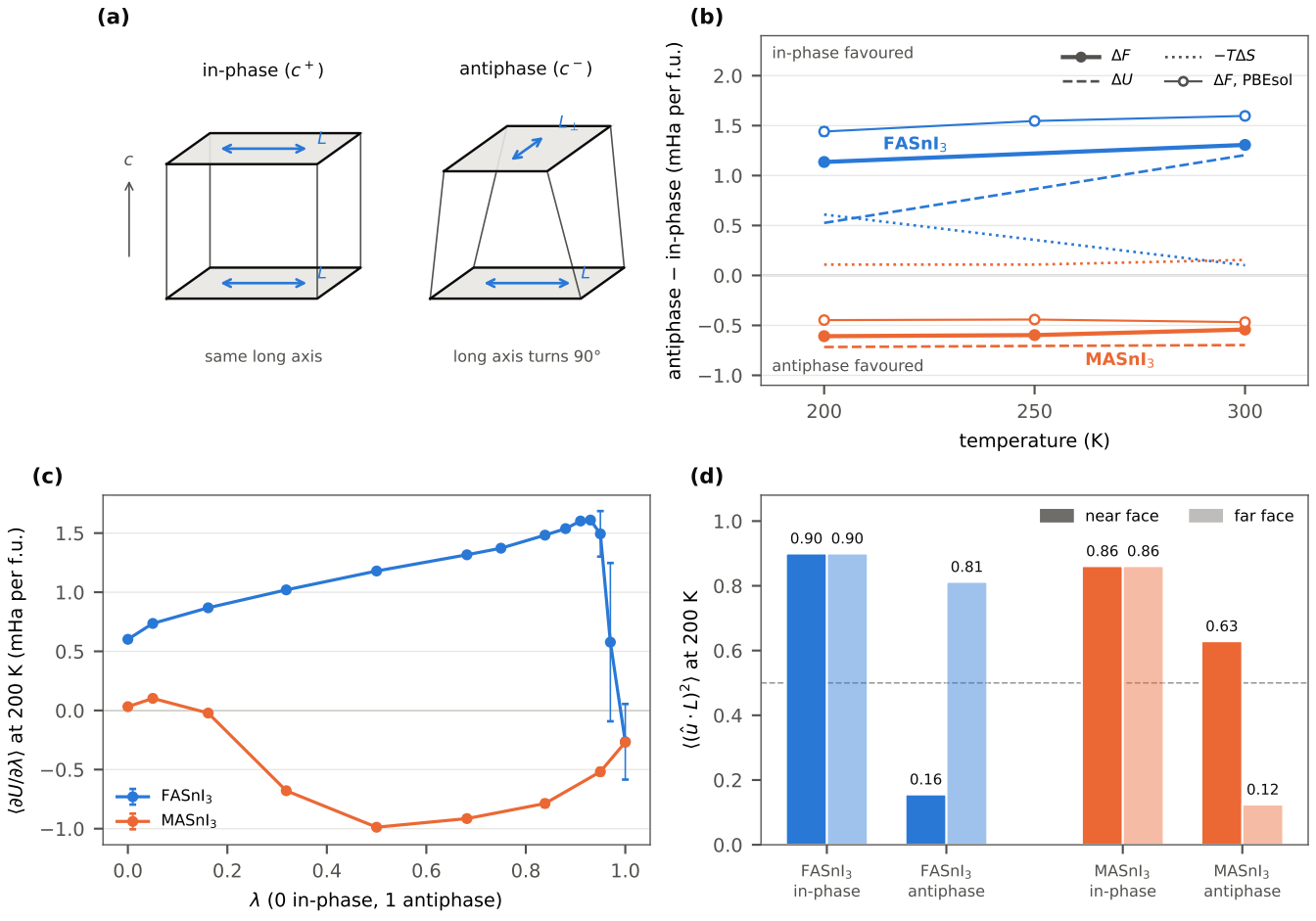
(a)
in-phase (c^+)
antiphase (c^-)
c
L
L
L
L⊥
L
same long axis
long axis turns 90°
(b)
in-phase favoured
antiphase favoured
ΔF
ΔU
−TΔS
ΔF, PBEsol
FASnI3
MASnI3
antiphase − in-phase (mHa per f.u.)
temperature (K)
(c)
⟨∂U/∂λ⟩ at 200 K (mHa per f.u.)
FASnI3
MASnI3
λ (0 in-phase, 1 antiphase)
(d)
near face
far face
⟨(û · L)²⟩ at 200 K
0.90
0.90
0.16
0.81
0.86
0.86
0.63
0.12
FASnI3 in-phase
FASnI3 antiphase
MASnI3 in-phase
MASnI3 antiphase